S1 Text. Supporting methods and numerical results

Electrostatic–hydrophobic cooperation in the mesoscale assembly of synapsin and synaptic vesicles

Shokoofeh Akbari, Oleg Shupliakov, and Yousef Jamali

S1. Design and structural observables

All numerical tables refer to the 12 trajectories in the factorial design: four E/H conditions and seed blocks 9310002, 9310003 and 9310004. Each run contains 200 SVs and 1,807 Syn agents in a spherical compartment of volume $0.1\,\mu\mathrm{m}^3$, evolved for 28 ms at a fixed integration step of 0.0007 ms. Van der Waals interactions are enabled in all conditions. Saved frames are separated by approximately 0.14 ms. Late-window quantities are averaged over frames 141–200 within each run (about 19.74–28 ms), then summarized across the three runs. Each run is one independent replicate. Reported decimal values are rounded only for presentation. Machine-readable full-precision tables are in S1 Data. All 12 trajectories ended with finite coordinates and residual geometric violations within 0.005 nm. The largest run-average fraction of updates affected by the drift limiter was 0.0030; per-run values appear below.

The association graph uses $\|x_i - x_j\| \le a_i + a_j + 2\,\mathrm{nm}$, where a_i is the outer radius. L is the all-agent largest-component fraction, Q_{Syn} is the non-isolated Syn fraction, and L_{SV} is the fraction of all SVs in the all-agent largest component. Non-isolated Syn includes Syn–Syn-only groups as well as Syn–SV associations. The underlying data column "synapsin_recruited_fraction" retains its historical machine-readable name and means exactly Q_{Syn}. The primary observable is L. The other two describe local protein participation and collective vesicle participation.

Extended E1H1 observation. The same three E1H1 trajectories were continued from saved checkpoints to 84 ms with unchanged physical parameters. The 2-nm association graph and saved-frame interval were retained; duplicate checkpoint frames were counted once. The extended trajectories are shown in Fig. 2C of the main text.

S2. Complete per-run structural observables

Condition	Seed	L	Q_{Syn}	L_{SV}	Drift-limited
E0H0	9310002	0.022272	0.305903	0.075750	0.000000
E0H0	9310003	0.023318	0.304409	0.077083	0.000000
E0H0	9310004	0.025677	0.297233	0.090833	0.000000
E0H1	9310002	0.091148	0.515643	0.168083	0.000642
E0H1	9310003	0.100797	0.519065	0.185000	0.000643
E0H1	9310004	0.104983	0.512249	0.198583	0.000615
E1H0	9310002	0.123559	0.699945	0.149333	0.000000
E1H0	9310003	0.082478	0.699613	0.102000	0.000000
E1H0	9310004	0.109841	0.697113	0.130667	0.000000
E1H1	9310002	0.551478	0.880419	0.622167	0.002962
E1H1	9310003	0.473194	0.882817	0.517417	0.002856
E1H1	9310004	0.633649	0.886054	0.693667	0.002984

The final column gives the whole-run mean fraction of particle updates affected by the deterministic drift limiter. Brownian increments are retained in full.

S3. Seed-block interaction and uncertainty

The primary interaction is $I_s = L_{11,s} - L_{10,s} - L_{01,s} + L_{00,s}$. It compares joint activation with the additive reference calculated from the other three conditions. All seed values are retained.

Seed	I_s
9310002	0.359043348
9310003	0.313237004
9310004	0.444502574

Mean primary interaction: 0.372261; between-seed SD: 0.066624; approximate 95% paired Student-t interval: [0.206759, 0.537763].

The main text reports paired differences as means and sample SDs. The additional statistical outputs below are retained for reproducibility. There are three paired blocks (two degrees of freedom). With eight possible sign-flip assignments, the minimum two-sided probability is 0.25 for nonzero effects. Paired parametric intervals assume approximately normal block differences, an assumption that is difficult to assess with three blocks. Full conditional contrasts and the additional statistical calculations are retained in S1 Data. The main comparisons use run-level means, between-seed SDs and paired seed differences.

S4. Association-shell sensitivity

Only the graph threshold changes; the trajectories and forces do not. Each entry below is a mean of three per-run late-window means.

Condition	Shell (nm)	L	L_{SV}
E0H0	0	0.013129	0.056028
E0H0	2	0.023756	0.081222
E0H0	4	0.043387	0.125833
E0H0	6	0.088078	0.219972
E0H1	0	0.056350	0.114972
E0H1	2	0.098976	0.183889
E0H1	4	0.150819	0.265944
E0H1	6	0.232697	0.391250
E1H0	0	0.052237	0.064750
E1H0	2	0.105293	0.127333
E1H0	4	0.163636	0.199056
E1H0	6	0.292047	0.356278
E1H1	0	0.423938	0.475000
E1H1	2	0.552774	0.611083
E1H1	4	0.656884	0.724083
E1H1	6	0.754243	0.830861

S5. Post hoc interface analysis

Let $\mathcal{S}$ and $\mathcal{V}$ be the Syn and SV index sets. For each Syn,

$$d_i^{\mathrm{SV}} = \sum_{j \in \mathcal{V}} \mathbf{1}\{\|x_i - x_j\| \le a_i + a_j + 2\,\mathrm{nm}\}.$$

The two Syn fractions count $d_i^{\mathrm{SV}} \ge 1$ and $d_i^{\mathrm{SV}} \ge 2$, respectively, divided by N_{Syn}. The reciprocal SV coverage counts SVs with any Syn neighbor, divided by N_{SV}. These fractions measure proximity. The analysis was developed after the simulations, using the same graph threshold for every condition.

Complete per-run late-window fractions are given below.

Condition	Seed	Syn: ≥ 1 SV	Syn: ≥ 2 SV	SV: ≥ 1 Syn
E0H0	9310002	0.180972	0.011502	0.808667
E0H0	9310003	0.184874	0.011538	0.807417
E0H0	9310004	0.175659	0.010035	0.784250
E0H1	9310002	0.399447	0.075669	0.958917
E0H1	9310003	0.397113	0.071168	0.958167
E0H1	9310004	0.384302	0.067829	0.957833
E1H0	9310002	0.682734	0.116676	0.998250
E1H0	9310003	0.681452	0.112000	0.997833
E1H0	9310004	0.679902	0.102195	0.995500
E1H1	9310002	0.874875	0.448275	0.999333
E1H1	9310003	0.877578	0.440814	0.998750
E1H1	9310004	0.879948	0.436709	0.999417

Complete unordered pair counts (mean number of edges per snapshot within each run):

Condition	Seed	Syn–Syn	Syn–SV	SV–SV
E0H0	9310002	150.050	348.400	109.083
E0H0	9310003	142.100	355.233	103.350
E0H0	9310004	142.733	335.750	107.367
E0H1	9310002	298.817	865.117	131.450
E0H1	9310003	300.417	852.300	128.950
E0H1	9310004	297.950	822.417	134.117
E1H0	9310002	203.950	1444.533	17.250
E1H0	9310003	203.733	1433.767	16.633
E1H0	9310004	195.100	1413.250	15.083
E1H1	9310002	889.250	2391.100	74.683
E1H1	9310003	880.550	2383.683	72.367
E1H1	9310004	876.517	2379.550	68.850

S6. Post hoc mobility and the experimental scale reference

Syn positions are projected onto xy, converted from nm to μm, and fitted using per-agent time-averaged MSD with a free intercept over native lags 2.10–3.92 ms (14 lag points). The reported apparent coefficient is slope/4 in $\mu m^2/s$. Negative fitted slopes are retained. The whole 200-frame trajectory is the principal scale diagnostic; the final 60-frame window provides a sensitivity check.

Condition	Seed	D: full	D: late	Negative: full	Negative: late
E0H0	9310002	0.140411	0.141157	0.0022	0.1007
E0H0	9310003	0.141014	0.138557	0.0017	0.1046
E0H0	9310004	0.139462	0.140225	0.0033	0.1184
E0H1	9310002	0.133545	0.130655	0.0033	0.1074
E0H1	9310003	0.134582	0.131626	0.0011	0.1051
E0H1	9310004	0.133554	0.132218	0.0050	0.1090
E1H0	9310002	0.113908	0.114040	0.0022	0.1129
E1H0	9310003	0.115999	0.112608	0.0022	0.1240
E1H0	9310004	0.115807	0.114354	0.0022	0.1140
E1H1	9310002	0.075550	0.066774	0.0077	0.1251
E1H1	9310003	0.076173	0.066835	0.0066	0.1522
E1H1	9310004	0.075010	0.064241	0.0089	0.1500

Hoffmann et al. (Nature Communications 2023, DOI 10.1038/s41467-023-42372-6) provide a Fig. 2h short-time mobility scale: source-data means are 0.0189192370 with SV and 0.0253090377 without SV, in $\mu m^2/s$. Source data: `https://doi.org/10.6084/m9.figshare.24179445.v1`. Their experiment selects condensate-core tracks at a 1-ms cadence, whereas the model analysis includes all Syn at a 0.14-ms saved cadence. Simulation positions have no localization error or exposure blur, and every model condition contains SV. Effective friction was chosen with reference to the literature before this comparison. The comparison provides a mobility scale reference, with higher absolute mobility in the model. Experimental uncertainty was not recalculated.

S7. Reproduction and data interpretation

S1 Data contains full-precision numerical inputs for the four main figures, structural tables and post hoc diagnostics, plus an inventory with SHA-256 checksums. The article-specific code package contains standalone extraction, statistics, rendering and supporting-table scripts. Published input trajectories are enumerated in its availability manifest; original raw simulation outputs are preserved separately. No trajectory is discarded or modified by the post hoc diagnostics.

Software validation. Graph extraction was checked against all-pairs enumeration on constructed and random geometries and on 24 saved frames. Structural observables and paired statistical calculations matched the released tables to absolute tolerance 10^{-12}. Mobility-estimator tests and SHA-256 checks also passed. Detailed cases and results are documented in `docs/ARTICLE_DIAGNOSTICS.md` in the reproducibility package.